

Enabling Effective Human-Robot Collaboration through Efficient On-Board Vision-Language Models

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Abstract—Vision-language models (VLMs) have recently demonstrated strong capabilities in extracting semantic and contextual information from the environment. However, current VLMs are typically designed for large-scale computing infrastructures and often rely on cloud-based inference, hindering their deployment on robotic platforms and raising potential privacy concerns when processing human data. Although smaller models are more suitable for on-board execution, they generally provide limited semantic capabilities and are often optimized for tasks such as visual question answering or image captioning rather than complex interaction scenarios. In this work, we propose a VLM pipeline for human interaction availability estimation, specifically adapted for on-device inference in collaborative robotic systems. The model enables robots to identify suitable interaction partners from direct visual observations by interpreting human activities and contextual social cues. We quantitatively evaluate the proposed approach in realistic indoor environments involving both single and multi-person scenarios. We further benchmark the end-to-end perception pipeline on an NVIDIA Jetson AGX Thor edge platform to assess its computational efficiency and suitability for on-board robotic deployment.

Index Terms—Vision Language Model, Edge Artificial Intelligence, Human Robot Interaction, Multimodal Perception

I. INTRODUCTION

Recent advances in Vision-Language Models (VLMs) have enabled richer semantic reasoning and high-level decision support for robotic applications [1]. However, their deployment on real robotic platforms remains challenging because of their computational and memory requirements. Consequently, many existing approaches rely on remote inference, reducing robot autonomy and introducing privacy [2], connectivity and operational concerns. Small VLMs are more suitable for on-board execution, but they are commonly designed for general-purpose tasks such as visual question answering and image captioning rather than socially grounded robotic reasoning [3]. Furthermore, the translation of semantic model outputs into actionable robotic behaviours is often only partially addressed [4]. Collaborative robots operating in shared environments must interpret socially relevant cues beyond geometric navigation and obstacle avoidance [5]–[7]. In particular, assessing whether a person is available for interaction is essential for

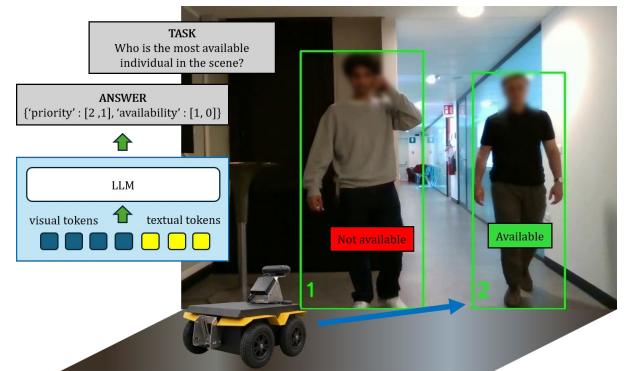


Fig. 1. On-device VLM for human availability estimation.

selecting appropriate partners, regulating engagement timing and avoiding socially inappropriate behaviours [8]. This becomes especially challenging in multi-person environments, where several potential interaction partners must be evaluated simultaneously. Given an RGB observation containing one or more detected individuals, the proposed system predicts a binary interaction-availability label for each person and a priority ranking over the detected candidates. These outputs support the robot in determining whether a suitable interaction partner is present and, when multiple candidates exist, whom to approach first. This provides the pre-interaction candidate-selection step required for proactive HRI.

The main contributions of this work are:

- To the best of our knowledge, we present one of the first on-device VLM pipelines specifically designed for human interaction availability estimation in multi-person HRI scenarios.
- We investigate a lightweight task-relevant token weighting strategy for structured VLM fine-tuning. The strategy increases the contribution of person identifiers and availability labels, which directly determine the downstream robotic decision.
- We quantitatively evaluate the proposed system on a cus-

tom human availability dataset, enabling experimentation in realistic indoor HRI scenarios. Furthermore, we define five representative interaction scenarios to systematically assess the model’s ability to infer human availability under different social conditions.

- We benchmark the complete perception pipeline on an NVIDIA Jetson AGX Thor, assessing its computational efficiency and feasibility for on-device robotic deployment.

The optimized model achieves a 28.53% relative F1-score improvement over standard fine-tuning and performance comparable to larger closed-source models evaluated in few-shot mode. On the NVIDIA Jetson AGX Thor, restricting generation to the required JSON output provides up to a $3.19\times$ inference speedup, demonstrating the feasibility of efficient on-board deployment.

II. RELATED WORKS

Recent studies have explored the use of VLMs in human-centered robotic applications, where robots must detect people, interpret social cues, and reason about human behaviour to support socially appropriate interaction and navigation [2], [4], [9], [10]. Among the most closely related systems, SocialLLaVA [4] enables on-board vision-language reasoning for socially aware navigation, but its reliance on long, unstructured outputs requires the generation of many tokens, making autoregressive decoding a major latency bottleneck on resource-constrained edge devices [11]. Focusing more specifically on the HRI literature, most existing engagement estimation methods assess whether a person is engaged during an ongoing interaction, rather than whether they can be proactively approached to initiate one [12]–[14]. Moreover, these approaches mainly consider single-person scenarios and show limited generalization to complex social environments involving multiple individuals [15], [16]. Their training data and annotation schemes are therefore centered on engagement maintenance and differ from those required by the proposed interaction-availability task. A further limitation is the absence of datasets explicitly designed for availability estimation in socially dense, multi-person settings. Existing resources generally involve few participants and do not provide balanced annotations of interaction availability [17]–[19]. For this reason, we evaluated the proposed model on a custom dataset specifically developed for this task.

III. METHODOLOGY

A. Interaction scenarios and social norms

We define five representative indoor HRI scenarios: (i) *single person*, where the model determines whether the only visible individual is available for interaction; (ii) *single available candidate*, where only one person in a multi-person scene satisfies the availability criteria and must be selected; (iii) *multiple available candidates*, where several individuals are available and must be ranked according to their suitability for interaction; (iv) *group interaction*, where people engaged with one another are identified as socially occupied and therefore

less suitable to approach; (v) *no available candidates*, where all individuals are unavailable and are ranked using secondary cues to identify the least disruptive interaction option. Availability is assessed from orientation, attention, ongoing activity, object use, interpersonal proximity, and group configuration [6]. Available individuals are prioritized, while ties and non-available candidates are ordered according to orientation toward the robot and proximity.

B. Data collection strategy

The dataset was designed to evaluate human interaction readiness in realistic indoor robotic scenarios. Data collection focused on recurring social situations in which a robot must decide whether to initiate an interaction, emphasizing visual cues related to social availability. The training dataset consists of 36 indoor recording sequences involving 27 participants, from which approximately 2,000 images were extracted at a sampling rate of 0.5–1 fps. During each recording, participants followed predefined instructions provided by the research team, specifying changes in posture, orientation, activity, interpersonal distance and social configuration. These instructions were designed to introduce meaningful variability across consecutive frames and to prevent the dataset from containing long sequences of nearly identical observations. Following [20], people detected by YOLOv11x [21] were represented through bounding boxes and numerical IDs, enabling multi-person grounding. In contrast, the validation and test sets were collected under more naturalistic interaction settings. The test set comprises 30 recording sequences, corresponding to approximately 600 images, with six sequences for each interaction category, while the validation set includes 15 scenarios, corresponding to approximately 300 images, with three scenarios per category. Both sets are fully separated from the training data and involve 16 participants who did not appear in the training set. Data were collected across both previously seen and unseen environments and under varying lighting conditions to assess the model’s generalization to unseen participants and diverse environmental conditions. The experimental setup and collection procedure were agreed upon with and formally approved by the Ethics Committee of Politecnico di Torino.

C. Image labeling and output format

Training and evaluation annotations were generated using separate protocols. Training labels are obtained via a teacher-guided annotation procedure, where a large multimodal model (*Gemini-2.5-Flash* [22]) generates initial availability annotations under explicitly defined task constraints. All teacher-generated annotations were subsequently reviewed frame-by-frame by a human annotator and manually corrected when necessary. This process corrected 8.55% of the person-level availability labels and 21.93% of the priority rankings in multi-person frames. In contrast, the validation and test sets were annotated entirely by two independent human annotators, without relying on teacher-generated labels. A third annotator

TABLE I
QUANTITATIVE COMPARISON OF THE EVALUATED MODELS ACROSS ALL METRICS.

Models	Acc. (%)	Prec. (%)	Rec. (%)	Spec. (%)	F1 (%)	$\bar{\tau}$	Top1-acc.	FIR(%)
Gemini-2.5-Flash	97.59	95.89	96.59	97.47	96.24	0.7342	91.42	4.01
GPT-5-mini	95.75	90.92	98.28	94.07	94.46	0.8491	95.88	10.58
FlowAct IAB	65.74	83.01	10.65	98.66	18.88	0.5125	79.75	1.82
HRI-NaVILA-base	82.28	91.08	57.31	96.59	70.35	0.6150	87.47	4.00
HRI-NaVILA	94.13	93.24	87.77	96.19	90.42	0.7342	90.39	4.38

TABLE II
PERFORMANCE ACROSS INTERACTION CONDITIONS FOR ALL EVALUATED MODELS.

Interaction conditions	Model	Acc. (%)	Prec. (%)	Rec. (%)	Spec. (%)	F1 (%)	$\bar{\tau}$	Top1-acc.	FIR(%)
Single person	Gemini-2.5-Flash	98.90	96.61	96.61	98.40	96.61	-	99.45	0.80
	GPT-5-mini	92.93	81.42	98.27	89.68	89.06	-	99.45	9.60
	FlowAct IAB	70.65	54.54	10.00	96.03	16.90	-	98.90	1.60
	HRI-NaVILA	95.35	81.66	84.48	91.91	83.05	-	100.00	0
Single available candidate	Gemini-2.5-Flash	97.70	96.39	97.27	96.33	96.83	0.9633	98.16	0
	GPT-5-mini	96.33	93.91	99.08	93.57	96.42	0.9816	99.08	0
	FlowAct IAB	58.98	95.45	19.09	99.07	31.81	0.8899	94.49	0
	HRI-NaVILA	98.62	99.07	98.16	99.08	98.62	1.0	100.00	0
Multiple available candidates	Gemini-2.5-Flash	96.00	99.47	95.91	96.55	97.66	0.7687	85.71	0
	GPT-5-mini	98.22	99.48	98.46	96.55	98.97	1.0	100.00	0
	FlowAct IAB	20.44	100.00	8.67	100.00	15.96	0.5510	72.44	0
	HRI-NaVILA	84.44	100.00	82.14	100.00	90.19	0.6122	75.51	0
Group interaction	Gemini-2.5-Flash	97.42	88.23	100.00	96.80	93.75	0.6513	86.20	13.95
	GPT-5-mini	97.85	93.47	95.55	98.40	94.50	0.6743	88.50	2.32
	FlowAct IAB	80.68	0	0	100.00	0	0.0689	35.63	0
	HRI-NaVILA	96.56	89.36	93.33	97.34	91.30	0.5823	80.45	6.97
No available candidates	Gemini-2.5-Flash	98.19	-	-	98.19	-	0.5345	80.18	3.77
	GPT-5-mini	92.82	-	-	92.82	-	0.7169	88.67	15.09
	FlowAct IAB	98.64	-	-	98.64	-	0.4528	74.52	2.83
	HRI-NaVILA	95.94	-	-	95.94	-	0.6981	84.90	8.49

subsequently resolved disagreements between the two annotators to obtain the final reference annotations. This separation limits the direct influence of teacher-model bias on model selection and final performance evaluation. The target output is a structured dictionary containing a priority ranking of the detected individuals and binary availability labels. By avoiding explicit reasoning generation during the decoding process, the model reduces inference overhead and improves inference latency, as discussed in Subsection IV-B.

D. Model description and fine-tuning strategy

The proposed approach builds upon a lightweight vision-language architecture (*NVILA-Lite-2B* [23]), consisting of a visual encoder, a language model and a multimodal projection module. The language model is adapted through QLoRA [24], while only normalization layers are updated in the vision tower, following [23]. Training uses a language-model learning rate of 2×10^{-4} , with the vision-tower rate scaled by a factor of $5 \times$. A batch size of one is used to remain within the 24 GB memory limit of an NVIDIA GeForce RTX 3090, while fine-tuning is limited to three epochs to mitigate overfitting and catastrophic forgetting. The fine-tuning objective was adapted

to the structured output format of the model. Each generated sequence follows a fixed JSON format, while the task-relevant information is encoded by person-ID tokens in the priority list and binary availability labels. Since standard cross-entropy assigns equal importance to all token predictions, we introduce the task-relevant Token Weighted Cross-Entropy (TWCE) loss to increase the contribution of errors affecting these numerical values:

$$\mathcal{L}_{\text{TWCE}} = - \sum_{w=1}^{|S|} \sum_{e=1}^{|V|} (1 + (\lambda - 1) \mathbf{1}_{\text{num}}(e)) \cdot y_{w,e} \log(\hat{y}_{w,e}) \quad (1)$$

where $|S|$ and $|V|$ indicate the lengths of the output sequence and the vocabulary, respectively, $y_{w,e}$ encodes the reference token at position w , and $\hat{y}_{w,e}$ represents the model's estimated probability for generating token e . The function $\mathbf{1}_{\text{num}}(e)$ activates only for numerical tokens, while the hyperparameter $\lambda > 1$ determines the degree to which prediction errors on numerical values are penalized during training. The hyperparameter was selected through grid search, with $\lambda = 15$ yielding the highest observed F1-score over the validation set.

TABLE III
AVERAGE LATENCY AND STANDARD DEVIATION OF THE *HRI-NaVILA* PIPELINE ACROSS DIFFERENT HARDWARE PLATFORMS.

Boards	Restricted generation (s)			Full generation (s)		
	Detection	VLM	Full pipeline	Detection	VLM	Full pipeline
Jetson AGX Thor 128GB	0.035±0.079	1.084±0.118	1.119±0.154	0.034±0.079	3.507±0.965	3.541±0.980
GeForce RTX 3090	0.029±0.032	0.817±0.119	0.846±0.121	0.029±0.032	2.493±0.901	2.522±0.900

This weighting strategy guides the model toward more accurate numerical predictions, resulting in improved quantitative accuracy.

IV. RESULTS

As discussed in Section II, no established benchmark is specifically designed for human interaction availability estimation in multi-person HRI settings. We therefore evaluate the proposed approach on the custom validation and test sets described in Subsection III-B. The validation set was used for hyperparameter selection, while all results reported below refer to the test set.

A. Metrics and evaluation

Availability estimation was evaluated using standard classification metrics, while priority ranking was assessed using the mean Kendall’s τ correlation between predicted and ground-truth rankings. To better characterize the downstream robotic decision at the scene level, we additionally report Top-1 accuracy, measuring whether the highest-priority candidate is correctly selected, and the False Interaction Rate (FIR), quantifying the proportion of scenes in which the system would initiate an interaction with an unsuitable candidate. Table I compares *HRI-NaVILA-base*, obtained through standard fine-tuning, with *HRI-NaVILA*, which incorporates the proposed task-relevant token weighting. The comparison also includes *Gemini-2.5-Flash* [22] and *GPT-5-mini* [25], evaluated in few-shot mode as large-scale VLM baselines [3]. Among the two proposed variants, *HRI-NaVILA* achieves the best overall performance, yielding a 28.53% relative F1-score improvement over the baseline. To further contextualize these results against a task-specific HRI approach, we additionally evaluate FlowAct IAB (Interaction Acceptance Belief) [15]. The method relies on a recurrent architecture based on Gated Recurrent Units (GRUs) to estimate interaction availability primarily from gaze direction and head pose. Body keypoints are then used to associate the estimated interaction state with the corresponding person in the scene. This represents a substantially different perceptual strategy from the proposed VLM-based approach, which considers broader visual and contextual information. In our setting, relying primarily on gaze and head pose is particularly challenging, as individuals may be distant from the camera, oriented away from it, or moving within the scene, making these cues difficult to estimate consistently. Moreover, the baseline does not explicitly model contextual cues such as object use, which are relevant to our availability definition. FlowAct IAB was evaluated using

its original pretrained configuration, since adapting the model to our dataset would require additional annotations that are not included in our protocol. The results therefore highlight the importance of richer semantic representations when interaction availability depends on contextual information that cannot be captured from gaze and pose alone. In particular, identifying social configurations such as groups requires reasoning about relationships among multiple individuals and their surrounding context. The lower performance of FlowAct IAB across the considered interaction categories further suggests that approaches relying on a restricted set of perceptual cues may struggle to generalize to more diverse social settings. These results motivate the use of VLM-based approaches, which can jointly integrate human activities, social configurations, object-related cues, and broader contextual information when estimating interaction availability. The category-wise results reported in Table II further illustrate how performance varies with the complexity of the social configuration. While performance generally decreases in more challenging multi-person settings, *HRI-NaVILA* maintains strong availability estimation across the considered categories and remains particularly effective in all interaction scenarios considered.

B. Latency performance

Latency was evaluated on the test set by restricting generation to the JSON output required for downstream decision-making. Measurements were collected on an NVIDIA GeForce RTX 3090 and NVIDIA Jetson AGX Thor. Table III reports the resulting device-dependent inference times. On the Jetson AGX Thor, the end-to-end pipeline latency decreased from approximately 3.541s with full-sequence generation to 1.119s when generation was terminated after producing the required JSON output, corresponding to an overall 3.19× speedup. The resulting latency is compatible with high-level human-robot collaborative tasks that do not require real-time control.

V. CONCLUSIONS

This work presented a VLM-based system for estimating human interaction availability in multi-person scenarios. Future work will address real-robot deployment, validation through questionnaires with human participants and a more comprehensive evaluation across a wider range of environments and individuals.

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